

Eodyne Systems

AI-Driven Clinical Decision Support for Stroke Telerehabilitation

The Problem

After discharge from inpatient care, stroke survivors enter a critical recovery window with limited clinical oversight. Conventional follow-up relies on periodic clinical assessments -- such as Fugl-Meyer scores -- that are infrequent, lagging indicators of recovery trajectory. By the time a plateau or regression is detected, the therapeutic window for intervention may have narrowed significantly.

Meanwhile, telerehabilitation platforms generate high-dimensional, continuous recovery data: session adherence, task performance, difficulty progression, kinematics, and patient-reported outcomes. Therapists managing large caseloads cannot routinely review this volume of data for every patient. Clinically meaningful patterns --early signs of disengagement, performance regression, pain-driven avoidance --go undetected.

Any decision support tool that aims to address this must be evidence-driven: every finding surfaced to the clinician must be traceable to specific, verifiable data.

Our Solution

We have built a clinical decision support system that continuously monitors patient recovery data from Eodyne's telerehabilitation platform and surfaces actionable, evidence-backed observations to clinicians --weekly, per patient, without requiring manual data review.

What it tracks:

- **Adherence.** Prescribed vs. completed rehabilitation minutes, per protocol and overall. Flags disengagement risk early.
- **Task Performance.** Success rates across therapeutic tasks over time. Detects regression, plateau, and under-challenge.
- **Difficulty Progression.** Adaptive difficulty modulation trends. Identifies overchallenge (rising difficulty with falling performance) and underchallenge.
- **Movement Quality Index (MQI).** A composite kinematic measure derived from stroke rehabilitation literature --incorporating movement smoothness, range of motion, and compensatory patterns --that is moderately predictive of Fugl-Meyer scores in clinical trials.
- **Patient-Reported Outcomes.** Pain, mood, and energy self-reports integrated alongside objective metrics to contextualize recovery patterns.
- **Patient-Protocol Fit.** Assessment-based matching (Fugl-Meyer, MoCA, PHQ-9, GAD-7) to evaluate whether assigned protocols remain appropriate.

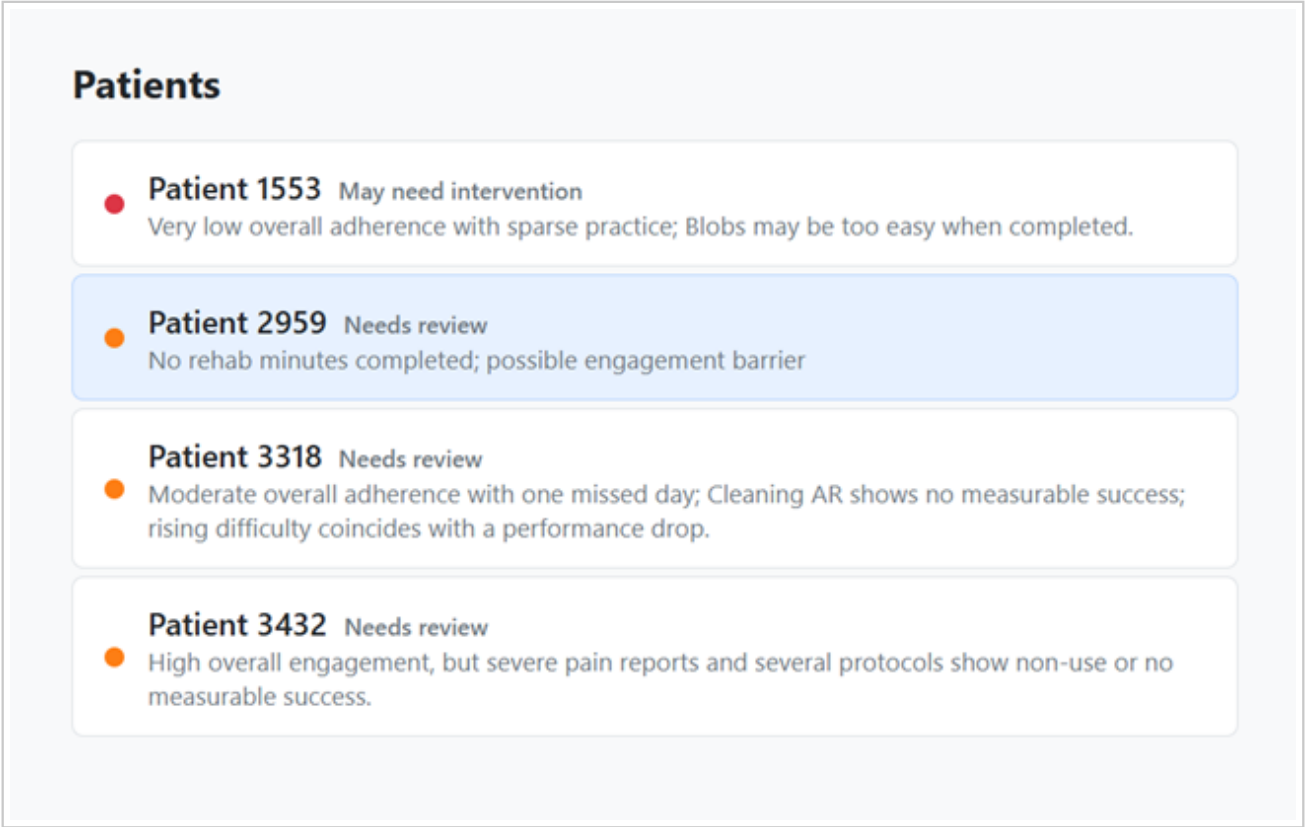
How it works:

The system ingests raw session, prescription, and assessment data weekly. A feature engineering pipeline computes clinically relevant signals from this data. A drift detection layer identifies patterns of concern: adherence drops, performance regression, overchallenge, fatigue cycles, and data anomalies. These signals are then synthesized by a medical LLM into structured, prioritized observations --each tagged with a severity level and linked to the specific evidence plots that support it.

Clinicians see a prioritized patient list with at-a-glance status. Selecting a patient reveals weekly observations, each clickable to surface the underlying time-series evidence. No finding is presented without traceable data.

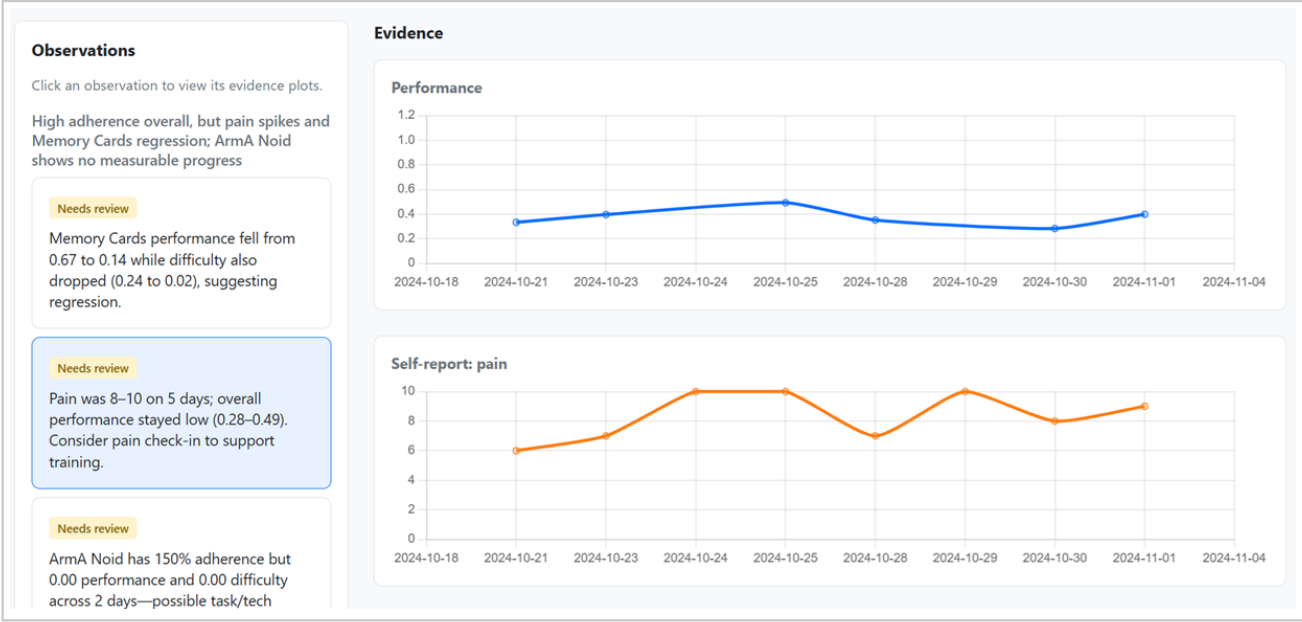
In Practice

The clinician dashboard provides an at-a-glance triage view across the caseload. Patients are ranked by attention level --from those on track to those who may need immediate intervention --with a concise headline summarizing the key concern.



Patient triage list with attention-level indicators and status headlines.

Selecting a patient reveals weekly observations generated by the system. Each observation is severity-graded and directly linked to the evidence that produced it. Clicking an observation surfaces the relevant time-series plots --performance trends, self-reported pain, adherence data --so the clinician can verify and contextualize each finding.



Observations with linked evidence: performance regression and pain self-reports.

Built on real clinical trial data from the NEST stroke telerehabilitation program. Designed for integration with Eodyne's rehabilitation platform.